

1. Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources over the internet. In GCP, this means you manage the infrastructure (VMs, storage, networking) while Google handles the physical hardware.

Key GCP IaaS Services

- **Compute Engine**
 - Virtual Machines (VMs) with customizable CPU, memory, and storage.
 - Predefined machine types or custom configurations.
- **Persistent Disk**
 - Block storage for VMs.
- **Cloud Storage**
 - Object storage for unstructured data.
- **Virtual Private Cloud (VPC)**
 - Networking services for secure communication.
- **Cloud Load Balancing**
 - Distributes traffic across multiple instances.

Use Case: Hosting applications, running custom workloads, lift-and-shift migrations.

2. Platform as a Service (PaaS)

PaaS abstracts infrastructure and provides a platform for building, deploying, and managing applications without worrying about servers.

Key GCP PaaS Services

- **App Engine**
 - Fully managed platform for web and mobile apps.
 - Supports multiple languages (Java, Python, Go, Node.js).
- **Cloud Functions**
 - Serverless compute for event-driven applications.
- **Cloud Run**
 - Deploy containerized applications without managing servers.

- **Firebase**
 - Backend services for mobile/web apps (authentication, database, hosting).

Use Case: Rapid application development, microservices, serverless apps.

3. Software as a Service (SaaS)

SaaS delivers ready-to-use applications over the internet. GCP offers SaaS solutions for productivity, analytics, and AI.

Key GCP SaaS Services

- **Google Workspace**
 - Gmail, Docs, Sheets, Drive for collaboration.
- **BigQuery**
 - Fully managed data warehouse for analytics.
- **Looker**
 - Business intelligence and data visualization.
- **Google Maps Platform**
 - Location-based services and APIs.
- **AI/ML APIs**
 - Vision API, Speech-to-Text, Translation API.

Use Case: Business productivity, data analytics, AI-powered applications.